

Read $f(2)$ and the preimages of 2.5.

Ans. $f(2) = 0$; preimages of 2.5: -3 and 3.

Expression used for the plot: $f(x) = 0.5x^2 - 2$.

5. Thread · Method [ALG] Program: choose a number; multiply it by -2; add 10; divide by 2. Result for 7? Expression in x?

   Ans. $\dfrac{-14+10}{2} = -2$; in general $\dfrac{-2x+10}{2} = -x+5$.

Sheet 74 Mon 31 May 2027 *False positives and false negatives*

1. Warm-up · Quick [PRB] Define "false positive" and "false negative".

   Ans. False positive: positive test for a person who does not have the disease. False negative: negative test for a person who has the disease.

2. Core · Method [PRB] 10 000 people, 2 % with the disease; the test is positive for 90 % of those with the disease and 10 % of those without. How many false negatives, false positives and positives in total?

   Ans. With the disease 200: 180 positive, so 20 false negatives; without the disease 9 800: 980 false positives; positives: $180 + 980 = 1\,160$.

3. Core + · Interpret [PRB] Using the previous data, calculate the proportion of people who truly have the disease among the positives and comment.

   Ans. $\dfrac{180}{1\,160} \approx 15.5\,\%$: since the disease is rare, most positives are false positives.

4. Review · True/false [PNP] True or false? Justify. The area of a square is proportional to the length of its side.

   Ans. False. $c = 1 \mapsto 1$ and $c = 2 \mapsto 4 \neq 2 \times 1$.

5. Thread · Disprove [LIT] Conjecture: "for every integer n, $n^2 + n + 41$ is prime". Test $n = 40$.

   Ans. $40^2 + 40 + 41 = 1681 = 41^2$: not prime. The conjecture is false.

   Successful tests prove nothing; one failed test disproves.

Consolidation, starter routines and transition to Year 12 (Première)

Sheet 75 Tue 1st June 2027 *Synthesis problems drawing on several topics*

1. Warm-up · Quick [EQ] Solve $3(x-1) = x+5$.

   Ans. $3x - 3 = x + 5$, $2x = 8$, $x = 4$.

2. Core · Method [GEO] For which value of k are the points $A(1;1)$, $B(3;5)$ and $C(k;9)$ collinear?

   Ans. $\overrightarrow{AB}\begin{pmatrix}2\\4\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}k-1\\8\end{pmatrix}$: $2 \times 8 - 4(k-1) = 0$, so $k = 5$.

3. Core + · Method [PRO] A price increases by 10 % then decreases by 10 %. What is the overall change?

   Ans. $1.1 \times 0.9 = 0.99$: a decrease of 1 %.

4. Review · Quick [PER] Total surface area of a cuboid with dimensions 4, 3, 2 (cm)?

   Ans. $2(4 \times 3 + 4 \times 2 + 3 \times 2) = 52$ cm^2.

5. Thread · Quick [SOL] Calculate the exact volume of a sphere of radius 3 cm.

   Ans. $V = \dfrac{4}{3}\pi \times 27 = 36\pi$ cm^3.

Sheet 76 Thu 3 June 2027 *Review of the six competences*

1. Warm-up · Metacognition [MET] List the six mathematical competences developed at lycée.

   Ans. Investigate, model, represent, reason, calculate, communicate.

2. Core · Metacognition [MET] Match each of the following to a competence: "translating a problem statement into an equation"; "sketching a curve"; "justifying an equality with a written calculation".

   Ans. Model; represent; reason (or communicate, depending on the emphasis placed on the write-up).

3. Core + · Metacognition [MET] To solve $2x + 3 = 11$, which competence is mainly used?

   Ans. Calculate (manipulating a linear equation).

   For oral discussion: setting up an equation from a problem comes under "model".

4. Review · Problem [MOD] Water at 20 °C is heated by 3 °C per minute: $T(t) = 20 + 3t$. After how many minutes does it reach 65 °C?

   Ans. $20 + 3t = 65 \Leftrightarrow t = 15$ min.

5. Thread · Quick [EQ] Solve in $\mathbb{R}$: $0.5x + 1.2 = 2.7$.

   Ans. $x = 3$

Sheet 77 Mon 7 June 2027 *Targeted reactivation of prerequisites for Première*

1. Warm-up · Quick [DEV] Expand $(x + 3)^2$.

   Ans. $x^2 + 6x + 9$.

2. Core · Method [EQ] Check that $x^2 - 5x + 6 = (x - 2)(x - 3)$ and deduce the solutions of $x^2 - 5x + 6 = 0$.

   Ans. $(x - 2)(x - 3) = x^2 - 5x + 6$; zero product: $x = 2$ or $x = 3$.

3. Core + · Anticipation [MOD] For $f(x) = x^2$, calculate the rate of change $\dfrac{f(3) - f(1)}{3 - 1}$.

   Ans. $\dfrac{9 - 1}{2} = 4$: this prepares the idea of the derivative in Première.

   Première topic, simple consolidation calculation.

4. Review · Construction [CON] Place M and N, the midpoints of $[AB]$ and $[AC]$, then draw $[MN]$. What can be said about (MN) and (BC)? Calculate MN if $BC = 9$ cm.

   Ans. $(MN) \parallel (BC)$ and $MN = \dfrac{9}{2} = 4.5$ cm.

5. Thread · Write-up [DEM] Prove that a triangle with sides 7, 24 and 25 is right-angled.

   Ans. $7^2 + 24^2 = 49 + 576 = 625 = 25^2$: converse of Pythagoras' theorem.

Sheet 78 Tue 8 June 2027 *Synthesis problems: choosing and justifying a strategy*

1. Warm-up · Reading code [ALG] What does $f(2)$ return?

```
def f(x):
    return x**2 - 3*x + 1
```

   Ans. $4 - 6 + 1 = -1$.

2. Core · Completion [ALG] Complete the code so that the function returns `True` when A, B, C are collinear:

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == ....
```

   Ans. We compare the determinant with 0: `== 0`.

3. Core + · Completion [ALG] Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        return x
    else:
        return ....
```

   Ans. `return -x`.

   Check: `vabs(-4)` *returns* 4.